



Impactful e-learning framework: A new hybrid form of education

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ABSTRACT

An e-learning model is helpful to enhance online connectivity and learning. This study aims to propose and develop an efficient e-learning framework to strengthen the online learning environment. The single consumer has to play a part in an e-learning system. There is a user-friendly interface for the proposed model. Architecture classifies into various categories. The work shows different parts of a successive model of e-learning and even each section's working process. This model and the structure can take all types of students into account. Since it is possible, there is a solution to this if the system grows using this method. Therefore, for teachers, it would be helpful too. The e-learning model facilitates self-learning for learners. Via this method, teachers can also do any operation. Using this form of system, they can also learn computer abilities and upgrade interactive skills.

Introduction

The conventional method of schooling is getting old every day. For better ease and simple ability growth, people use the Internet and various mobile gadgets. The e-learning method is distinctive. The learning methodology should be well structured to execute a diagram or GUI. This survey includes an interface to eLearning focused on the interaction to e-learning. The interfaces suggested several new features that consider both teacher's and students' experiences for learning systems success. When entering an e-learning system, time and unique theme-based learning are time-consuming. It is better to resolve the obstacles. The longitudinal research explores the efficiency of problem-based education in enhancing success in a computer program course. In addition to the numerous obstacles and inspired tasks, there were significant variations in the course's success. A user-friendly and quick to use GUI leaves users very comfortable and increases efficiency. A highly scalable e-learning model with strong management allows to manage the system and retain user details. The gathered data can be monitored to understand better how happiness is used. The research seeks to provide a graphical analysis of the features of the instructor's happiness in e-learning. Various academic and non-academic classes can be included through the e-learning model approach. For classrooms, colleges and universities, an e-learning mode may be used. With accredited classes,

people can develop their skills at academic and non-academic places. Three university courses have annotated the learning material, modifying and developing it. The research seeks to provide an instructional and learning process structure for a university. The proposed survey was divided among 297 participants to test the hypothesis. The paper introduces a multidisciplinary decision-making approach and is intended to help providers configure programs. A mixed learning network allows the Internet to merge conventional school styles with synchronous e-learning. Pilot research with the proposed combined learning framework indicates that underprivileged school students will access better education.

In this paper, the authors are seeking to propose and build an impactful e-learning platform. In academic formations, this model can reduce complexities and various problems encountered by both students and teachers. This model can be conducted for students, teachers, and administration in offline and online interaction methods due to considering network issues over urban and rural environments. This e-learning model facilitates self-learning for learners, emerging assessments to teachers, and overview trackers for administration. A cloud database is used to store various data from students and teachers. Track record of both students and teachers is stored in the cloud database system, where teachers can view students' profiles and records, and students can view their records to grow a progressive report.

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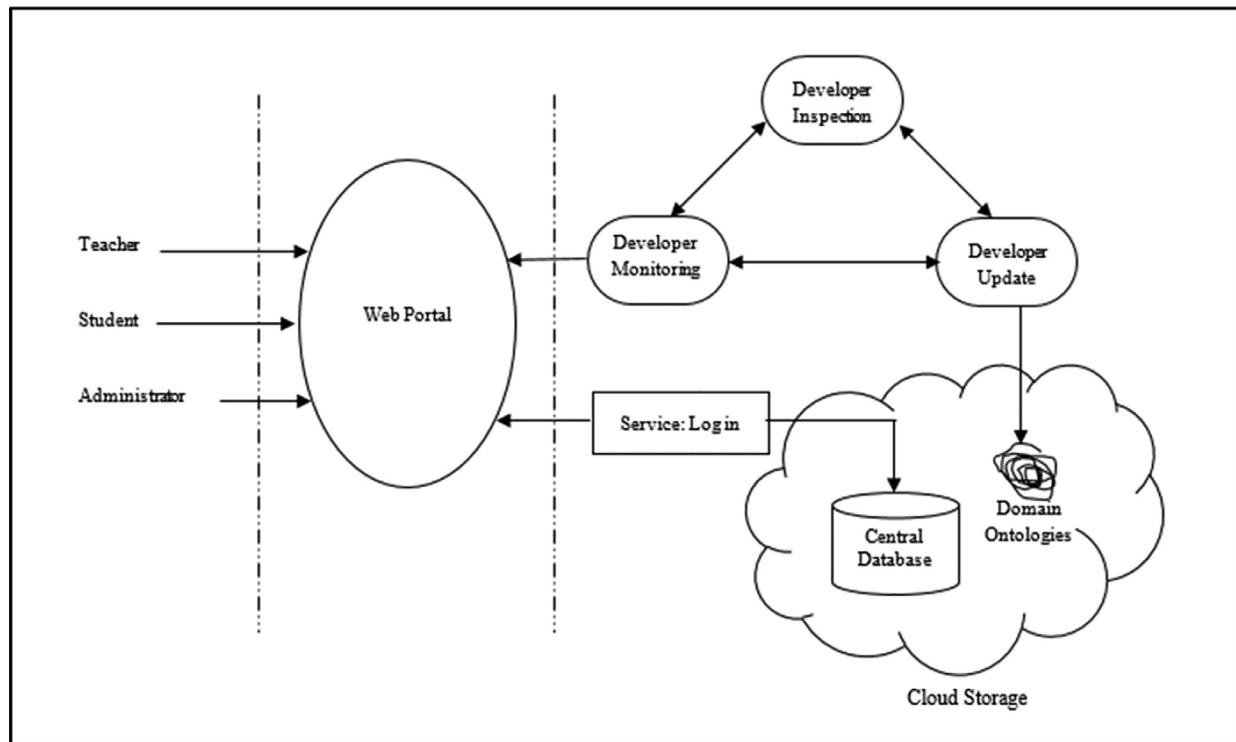


Fig. 1. The architecture of a cloud storage access model.

Literature review

Farhan et al. (2019) proposed and designed an e-learning user interface with the help of web programming languages, and also it was tested collecting quantitative data from teachers and students to ensure that how effective it is. Using qualitative and quantitative empirical approaches, students and instructors have more excellent knowledge of the scheme.

Yengin et al. (2011) In this analysis, the e-learning paradigm's effectiveness was submitted to teachers, e-learning designers and illustrated professional satisfactions utilizing positive and negative feedback on variables linked to the teacher's happiness in e-learning systems.

Sarwar et al. (2019) proposed a mathematical model for online learners using machine learning techniques. The learners of this method were divided into separate divisions. The learning material glossed through the ontology of classes, of which three academic classes (C++, C #, and JAVA) were represented.

Vasconcelos et al. (2019) presented decision-making methods based on multidisciplinary requirements and various criteria for the configuration of online course programs. The decision-maker configuration model developed, program institution-based mentors listed practices relevant to the consistency of service design implementation processes.

Tawafak et al. (2019) present in intention to establish a perspective to improve university education and learning and the dynamics of course evaluations by analyzing the relationship between the e-learning implementation variables and the paradigm for technology adoption, presenting a high degree of relevance and reliability outcomes.

Kardipah and Wibawa (2020) presented the evaluation for success and interview guidelines, a flipped-mixed learning paradigm for developing students' programming skills, and examined the use of problem-based learning in FBL student success inspired students to learn more.

Dey and Bandyopadhyay (2019) aim to conduct using the proposed blended learning platform to show that underprivileged school students' quality education can be provided.

From this idea, we are launching a proposal on an e-learning system based online and offline, both with a new blended learning module. We will reduce the complexity encountered by both students and teachers in online learning in emergency circumstances.

Experimental method

e-Learning is a process where we try to come out from the traditional learning system and enter the electronic learning system using different electronic devices. Online activities are now very much popular as it helps people to learn quickly and more conveniently. Delivering information and learning materials to the students is the primary approach of e-learning. The complete functionality of the model is divided into three parts.

Research model

This system will be used from all of the country's rural places so that the whole system will be used with the Internet through a website. The developers will make a website used by the Teachers, Students, and the Administrator. The developers can make changes to the websites and give updates to the website if it is needed to the domain. The Teachers, Students, and the Administrator can log in to their profile and using the service login function, and they can enter into the user Database (DB) and approach their functions. The architecture of a cloud storage access model is shown in Fig. 1.

User: Our proposed system will be used by different users like students, teachers, and administrators. They can access their particular sites. Each user will have a unique profile here. They can make changes to it if they want. Teachers can provide all necessary things here for the students by accessing the web portal. Students will get all learning materials here, and they will also be able to attempt different assessments. The administrator can monitor all things.

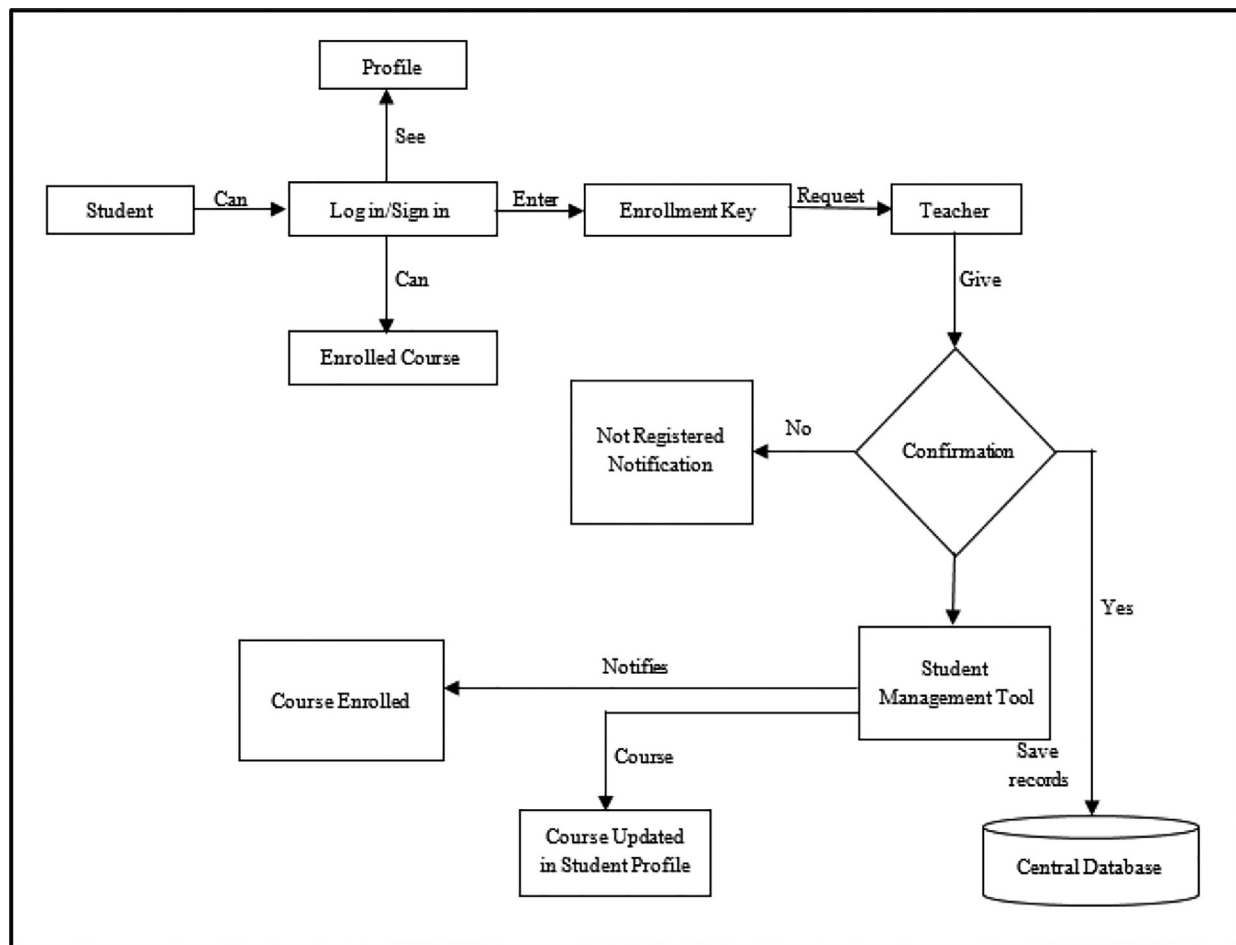


Fig. 2. Course enrolment.

Web portal: A web-based objective that collects information from various sources. Users can access the web portal. Users have to log in to this portal to use this system. If users have no account, then they will have to create a new account. It is the website of the e-learning system. Only the developers can make changes to this website because developers will create this site. The web portal will be updated when the developers will find new features to update this site.

Central database: In the database, all performances will be recorded automatically, like student's or teacher's profile information, student's quiz marks, assignments marks, mid or final exam marks. It is needed to make the results of each student. Otherwise, all of the essential data can be lost or deleted. From the database, the administrator can also see the performance of teachers.

Course enrolment

First, a student needs to create an account and create a profile with valid information. To enroll in a course at first, a student needs to register for that particular course. Then an enrolment key will be provided to him. Entering the key, they will be enrolled in the course, and the course information will be enlisted into his profile. If the student enters a wrong enrolment key, he will be notified with a notification that will show "Not Registered." Here the whole system will be monitored by a student management tool. The complete course enrolment system is shown in Fig. 2.

Teacher's activity

A teacher's activity is a process for all the tutors to manage their courses, providing links, questions, discussion forums, quizzes, assignments, and exams. First, they need to create an account in this e-learning system. Then they can create courses. After creating an account, all activities will be monitored by administrator system management through the teacher's management tool.

The teacher's activity diagram is shown below in Fig. 3.

Administrator

The administrator can control the student management system, teacher's management tool, course management tool, and manage the whole system's security.

The student management tool contains different tools.

- 1) Manage department is a sector that handles all the departments. The as different department has a different number of students and teachers.
- 2) Management section is almost the same as the manage department. It deals with the different sections that are present in a department.
- 3) Manage class (with limited student) is a section that deals with the limited numbers of students presented in that particular class. It helps to organize the whole approach of the laves of students.

Finally, record student performance is a sector that deals with the activity of all the students. How they are working, their performance,

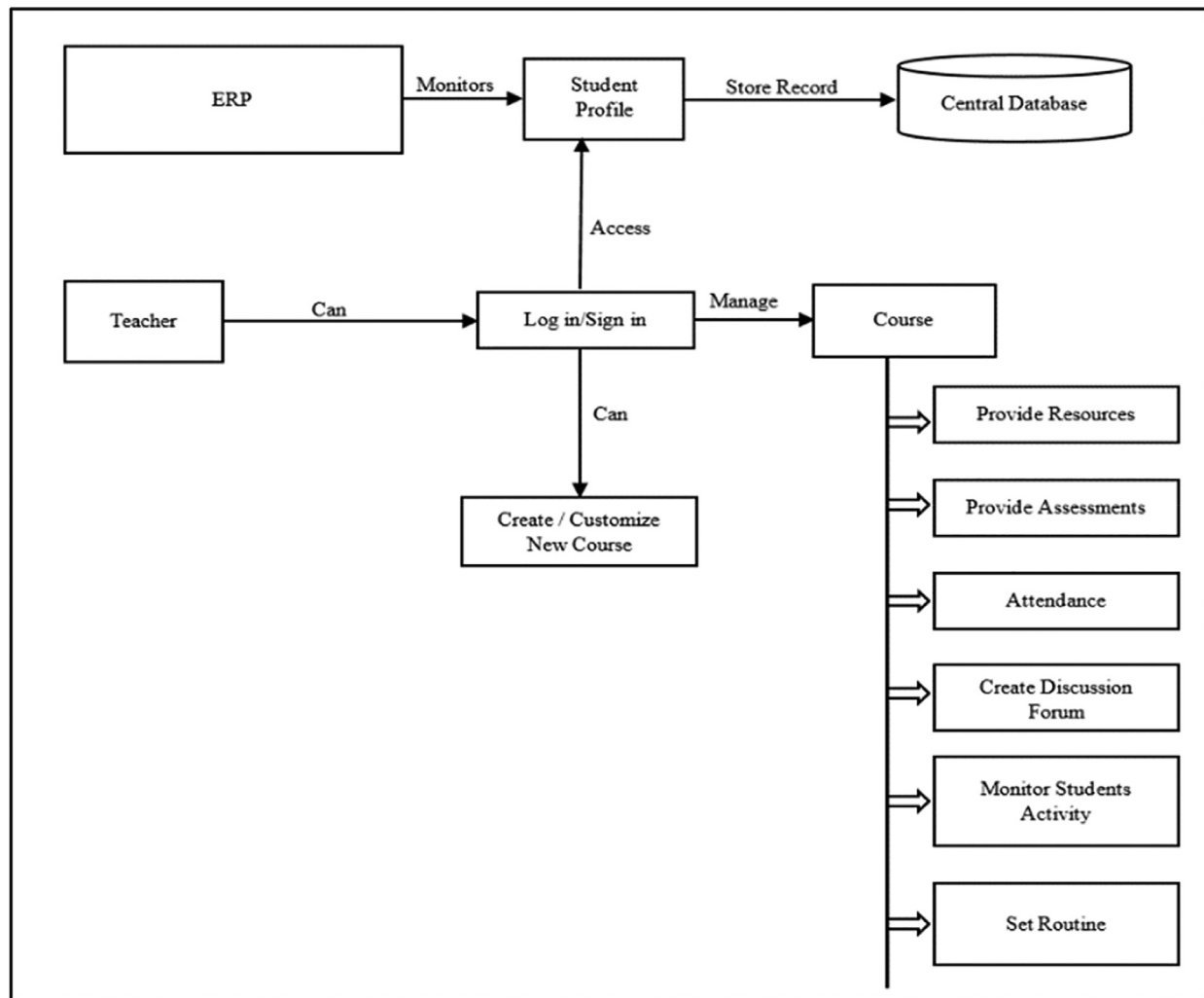


Fig. 3. Teacher activity.

and the students' feedback for the course and the teachers. The administrator segment has been cleared by the figure given below in Fig. 4.

Exam and assessments

Online or offline interactive classes, exams, or assessments have been taken with a strict schedule and curriculum. Thus, in this proposed model, Exam and Assessments is segmented into two types: Multiple Choice Questions and Written Exam.

Multiple choice question

For the MCQ (multiple choice questions), the quiz system is straightforward. The approach is that a teacher will provide several questions and set an answer sheet. When the student attends the quiz, they will only see the questions and answer these questions accordingly. After answering all the questions, the answer sheet of the student will go inside the system, and then the answers will be compared with the provided teacher's answer sheet. Marks will be enlisted into the Record Student profile and the database, and the result and the answer sheet will be shown to the students. The working process of MCQ based assessment is shown in Fig. 5.

Written assessments

For a written exam, there is a different approach. There will be a time limit in this kind of exam; the students need to submit the answer sheet.

If they cannot submit within a given time, there will be negative marking. Then the whole process will wait for the teacher's correction. Often, when attending tests or examinations, students face network issues. When students try tests, several more pages are automatically loaded, including the current one. If a network issue is encountered, a solution can be provided. If the Internet fault is detected, the previously loaded homepage will load the current database and appraisal pages. After the teacher's grading, the result will be enlisted into the Record Student Profile and Publishing the result where the students can see their marks. Furthermore, the whole dataset will be recorded into the database form of the Record Student Profile. The Written Exam or Assessment Module Diagram is given below in Fig. 6.

Technical architecture

Here is a use case diagram for the approach of the online base electrical learning system. A teacher can create courses with an enrolment key. The students can enroll in that course using that particular key. The administrator will monitor the whole course. The course contains a routine for the class. So, the routine can be accessed by the teachers, students, and the administrator. Here every learning material is provided by the teachers with the help of Provide Resource. All of them can be accessed by the students. Students can download the Resources and study offline. They will get all the assignments and all kinds of links through this method. Links will be provided to them with different drive access and different articles. Teachers can communicate with the students with

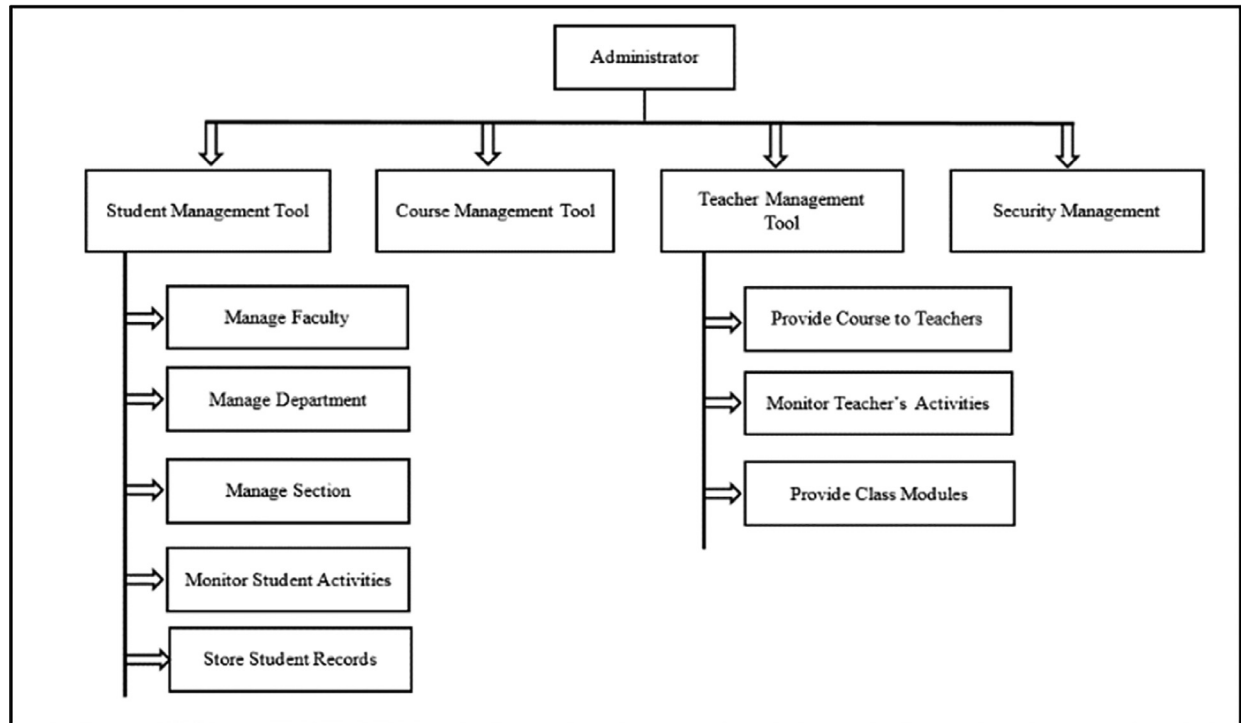


Fig. 4. Administrator activities.

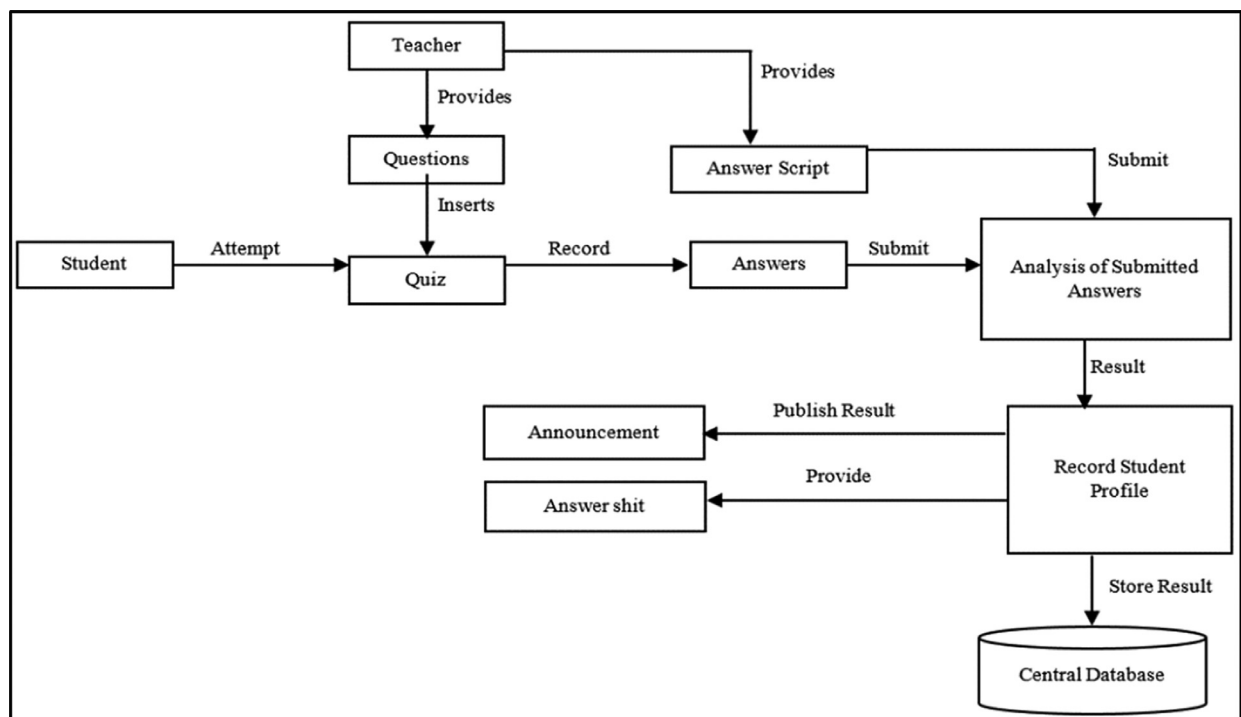


Fig. 5. MCQ based assessment.

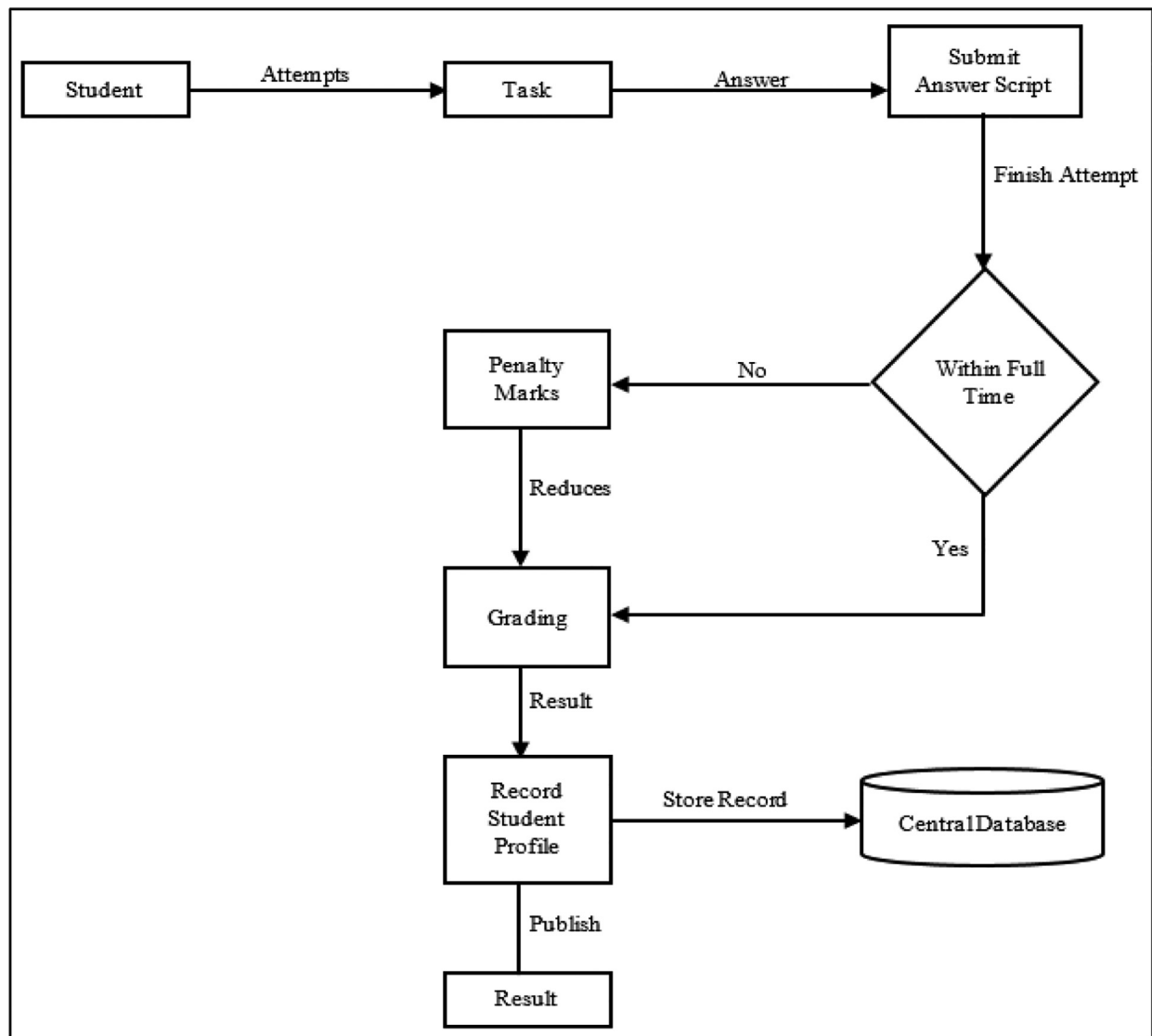


Fig. 6. Written exam assessment module diagram.

the help of Forum Communication. He can open different forums if he wants, and the students can discuss different subjects. They can also give feedback on the course through this forum. The discussion feedback needs monitoring. So, the administrator can also have access to this. The administrator will have access to the teacher's profile as the teacher's profile contains all the course feedback. The teachers will take the quiz. Here the teachers must include the questions, but he can either provide the students with an answering script. The result will contain the student's profile that can be accessed by the students, teachers, and the administrator. An e-learning system contains many courses. So, the administrator needs to maintain all the courses. He also needs to maintain the whole security system of the system (Fig. 7).

Perceived challenges

Students need to link to the academic course online and offline to have a hassle-free education system. It is complicated in Bangladesh to communicate with academic views online and offline because of network issues. The most challenging facts were the existing high Internet-based curriculum and typical form of assessment and education. Education System will become a comfort zone by reducing extra hassles caused by various blurry perceptions.

Conclusion and future work

One of the practical usability of a system is satisfaction. The provided model has a user-friendly interface for the users. It contains a clean and simple approach for an eLearning interface. It provides a simple guideline for a better learning system. This model is based on an offline study approach system and only an online exam system for proper judgment. The system has a dynamic approach for better productivity. The feedback on the model will help the developers to improve the model as it contains suggestions. This proposed model aims for a more holistic approach that takes the subsequent work into account.

- Implementation of the given proposed model.
- Connect both students and teachers' perspective.
- Visual Implementation with a virtual environment.

Our plan for linking urban areas and downtown to the whole system is being developed. The education system may need to be developed to make a stronger nation shortly. However, there needs to be more research. This work is a complete but step-by-step proposed model.

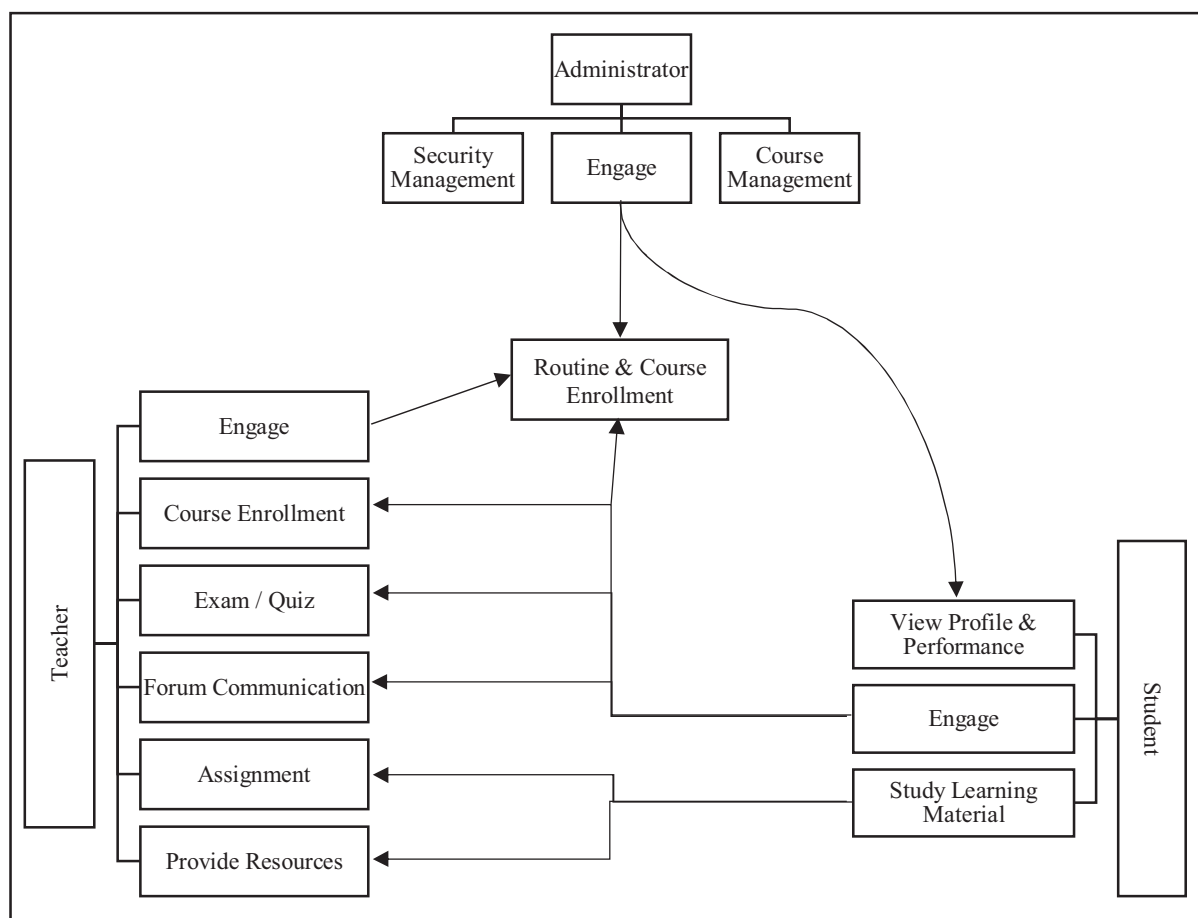


Fig. 7. Technical architecture of the proposed system.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No datasets are used for this research work.

Code availability

No codes are used for this research work.

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